TODO Title: Structure-Aware Document-Level EN↔VI Legal Translation with a Fine-Tuned 1.8B Model

Anonymous ACL submission

Abstract

We describe our system for the VLSP 2026 shared task on English↔Vietnamese legal machine translation, where whole Markdown documents are translated by a model of fewer than 3B parameters that runs locally without external APIs, and inference time and memory are also evaluated. We fine-tune Hy-MT2-1.8B with LoRA on segment-level legal data and run it inside a structure-aware document pipeline: Markdown is parsed into nodes, prose is packed into blocks without marker tokens, outputs are checked for lost headings, labels, uncovered source and unfinished generations, and failed units get one recovery pass with a changed input. On the 50-document public test, an earlier version of the pipeline gains +1.2 (EN→VI) and +1.4 (VI→EN) BLEU by packing neighbouring nodes into one request instead of translating nodes one by one. A second LoRA stage on our v3.3 data adds +1.15 BLEU EN→VI (95% interval excludes 0) and +0.67 VI→EN (interval includes 0); these are single runs, the A1 baseline ran under a different PyTorch version, and the two adapters score about equally on a held-out internal test. The submitted system reaches 70.87 / 47.44 BLEU on the public test; these scores may be optimistic because 72 terminology rows of v3.3 were taken from the public references.

1 Introduction

[TODO: Linh viết]

2 Related Work

Small LLMs for translation. Recent open translation models are decoder-only LLMs adapted to translation, from the two-stage fine-tuning of ALMA (Xu et al., 2024) and the continued pre-training and instruction tuning of Tower (Alves et al., 2024) to the Chinese- and English-centric NiuTrans.LMT suite (Luo et al., 2025) and the Hunyuan family (Zheng et al., 2025). We build on Hy-MT2, whose 1.8B variant covers 33 languages and is reported by its authors to compete with commercial APIs overall (Zheng et al., 2026). Encoder-decoder systems such as NLLB (NLLB Team, 2022) cover many languages but do not take instructions, so they cannot use prompt-based terminology hints.

English–Vietnamese translation. Large public English–Vietnamese resources are sentence-level corpora: PhoMT (Doan et al., 2021) (about 3M sentence pairs) and MTet (Ngo et al., 2022) (4.2M training pairs and a multi-domain test set), the latter with the EnViT5 models. They do not provide structured legal documents, so we build segment-level legal training data instead (Section 3).

Document-level and legal translation. Context-aware NMT showed that sentence-level systems are inconsistent in deixis, ellipsis and lexical cohesion (Voita et al., 2019), and surveys cover the resulting model families (Maruf et al., 2019); LLMs have been studied as document-level translators (Wang et al., 2023). Recent legal translation work adapts pretrained models to legal text, e.g. QLoRA fine-tuning of IndicTrans2 on the JUST-NLP 2025 legal corpus (Singh et al., 2025) and domain-specific fine-tuning of an open-weight LLM on EU legislation (Lorini et al., 2026). Our setting differs in that the unit is a whole Markdown document whose headings, article numbers and list labels must survive translation, so we treat structure preservation and failure recovery as part of the translation system rather than as post-editing. We also guard against degenerate repetition, a known failure of likelihood-based decoding (Holtzman et al., 2020), and use a repetition

penalty (Keskar et al., 2019) only in a retry path.

3 Data

[TODO: Linh viết (kèm hình pipeline xử lý dữ liệu nếu có)]

4 Method

4.1 Backbone and fine-tuning

We use Hy-MT2-1.8B (Zheng et al., 2026) (dense, fp16, about 5 GB of VRAM), which fits the 3B limit and one Kaggle T4. We also ran a pilot with NiuTrans LMT-60-1.7B (Luo et al., 2025): on 42 raw development documents the unmodified LMT model dropped from 50.3 to 9.9 BLEU (EN→VI) and from 31.4 to 15.4 (VI→EN) when given 1,194-token multi-paragraph chunks instead of single paragraphs. The two pilots used different data budgets (5.3k vs. 53.5k rows) and recipes, so they are not a controlled comparison and we do not rank the two models on this evidence.

Training is causal-LM cross-entropy on the assistant turn only (translation plus end-of-sequence); prompt tokens are masked, as in our training script. Each pair uses a "mixed" prompt: with probability 0.5, if glossary pairs (442 terms) occur in both source and target, a terminology-hint prompt with at most eight pairs is used, otherwise the default instruction; inference always uses the default (Appendix A). Stage A1 trains LoRA (Hu et al., 2021) (r=64, α=128, all attention and MLP projections, 4.15% of parameters) on 53.5k segment-level pairs (mainly article and clause segments) for one epoch at 2×10^{-4}. Stage v33 *warm-starts* from the A1 adapter and trains one more epoch on the 82,203 pairs of data v3.3 (Section 3) at the lower rate 5×10^{-5} (2×10^{-4} is the rate for training LoRA from scratch); details are in Appendix B.

4.2 Document pipeline

Figure 1 summarises the `docmt` pipeline, written as one module per step so that each step can be changed and measured alone.

Parse and pack. A rule-based parser turns a Markdown document into *nodes* (headings, article and clause labels, paragraphs, table cells, signature blocks) such that concatenating the nodes reproduces the source exactly. In an earlier version we compared three packings (Section 6): one node per request, up to 8 neighbouring nodes (at most 768 source tokens) joined with marker tokens, and the same grouping joined only by newlines. The submitted version ("legal prose blocks") first merges adjacent translatable prose nodes of the same scope (headings, articles, clauses, points, list items, paragraphs) into one block, and closes a block once it exceeds 768 source tokens or 8 nodes and its last node ends with a full stop. A new article or a new chapter or section heading starts a new block, a title chain may join the article that follows it, and tables and signature blocks stay separate units. Each block is one translation unit and its output is not required to map line by line onto the source nodes, so no line-count check is applied to prose; instead, every source span must be covered exactly once.

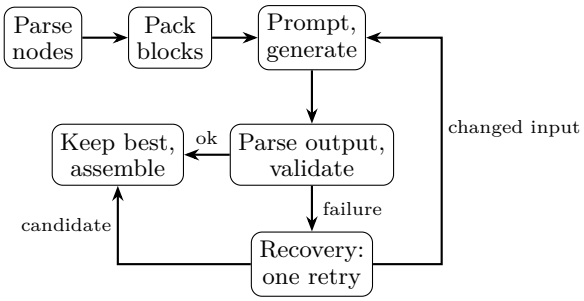


Figure 1: Document pipeline with one recovery round. At inference the pipeline reads only the source text.

Validate and recover. Each output is checked for source-span coverage, preserved headings and numbered labels (treating đ and `dd` as the same label), and unfinished generation (no end-of-sequence before the budget, or a detected repetition loop). A unit with such a failure gets *one* recovery round with a changed input. After a lost label the prompt adds an instruction to keep the source labels; after a VI→EN unit ran to its budget the earlier (legacy) budget is used; after a repetition loop the repetition penalty is 1.15 (Keskar et al., 2019). The best translation obtained is kept; a document with unresolved units is still output but flagged, and a span for which the model returns nothing keeps the source text, also flagged, so no content is silently dropped.

Decoding and runtime. Decoding is greedy in fp16. The output budget depends on direc-

tion, max(256, $\lceil 2.5s \rceil$+128) tokens for EN→VI prose and max(256, $\lceil 1.25s \rceil$+128) for VI→EN prose, where s is the source length; Vietnamese references need about 1.5× more tokens than English. Requests are batched by length to reduce padding, and the batch size (8 or 16) is calibrated on each GPU. In an earlier comparison with the same A1 model, data, budgets and batch size 8, scheduling requests by length cut the public-test wall time of the whole job from 12,473 s to 8,080 s (−35.2%), with BLEU changing by −0.01 (EN→VI) and +0.05 (VI→EN).

Anti-repetition. A GPU-side detector checks the generated tokens every 32 decoding steps and aborts a degenerate generation early; an aborted unit counts as unfinished and goes to the recovery round above (Holtzman et al., 2020). The first pass uses no repetition penalty. With the same v33 adapter and software environment (single runs), a first-pass penalty of 1.05 scored 70.28 / 47.40 BLEU and 1.0 scored 70.87 / 47.44, so we use no penalty in the first pass; the difference comes from one run per setting and has no confidence interval.

5 Experimental Setup

[TODO: Linh viết]

6 Results

[TODO: Linh viết]

7 Conclusion

[TODO: Linh viết]

Acknowledgments

[TODO: nhắc người thứ ba làm Hydra (nếu bài có nhắc tới) — hỏi Linh tên/cách ghi]

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A Prompts

Both prompts are the ones used in the A1 training script, sent as the user turn of the Hy-MT2 chat template; `{src}` is the unit text and `{tgt}` is "English" or "Vietnamese". Inference always uses the default prompt.

Default.

```
Translate the following text into {tgt}.
Note that you should only output the
translated result without any additional
explanation:\n\n{src}
```

With terminology (training only).

```
Reference the following
translations:\n{refs}\n\nTranslate the
following text into {tgt}.  Note that you
must ONLY output the translated result without
any additional explanation:\n\n{src}
```

Each line of `{refs}` is "*source term* translates to *target term*", with at most eight pairs.

B Hyper-parameters

Adapter checksum (SHA-256):
`30b5fc68a31c3de3d722a6dee5484aaf`
`98c4f6b9d7b41fb55ac70f6d23a3ba89`.
Base model revision: `9a341cd1b679d3ef`
`d23b46e847b01745a71ed792`.

LoRA fine-tuning (A1, then v33)	
Base model	Hy-MT2-1.8B (fixed revision)
Adapter	r=64, α=128, dropout 0.05
Target modules	q, k, v, o, gate, up, down
Trainable parameters	77.6M (4.15%)
Data	A1: 53.5k pairs; v33: 82,203
Learning rate	A1 2×10^{-4}; v33 5×10^{-5}
Schedule (v33)	cosine to 10%, 25 warm-up
Epochs	1 (v33: 2,569 steps)
Batch	$1 \times 16 \times 2$ GPUs $= 32$
Precision, max length	fp16 DDP, 3,072 tokens
Weight decay, clip	0.0, 1.0
Seed, hardware	42, Kaggle 2×T4
Inference (submitted setting)	
Decoding	greedy, fp16, penalty 1.0
Recovery	one round; penalty 1.15 after a loop
Prose block	closed after 768 tokens or 8 nodes, at a full stop
Output budget	EN→VI $2.5s$+128; VI→EN $1.25s$+128
Batch size	8 (calibrated from 8/16)
Context limit	8,192 tokens
Post-processing	none

Table 1: Training and inference settings. s is the source token count; budgets have a floor of 256.